

# Claudio César Claros Olivares

📍 Newark, DE, USA    ✉ cesar@udel.edu    🌐 cesar-claros.github.io    📱 cesar-claros    🌐 cesarclaros  
 🎓 Google Scholar    🆔 0000-0001-7671-9562

## Education

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>Ph.D. University of Delaware</b> , Electrical and Computer Engineering<br><ul style="list-style-type: none"> <li>• Advisor: Austin J. Brockmeier, Computational Neural and Information Engineering Lab</li> <li>• Research: reliable machine learning under distribution shift (out-of-distribution detection, calibration) and machine learning for brain imaging and sports medicine</li> <li>• Bendett Fellowship (2023); Best Poster Award, ECE Research Day (2024)</li> </ul> | Newark, DE, USA<br>Jan 2022 – present  |
| <b>M.Sc. University of Delaware</b> , Electrical and Computer Engineering<br><ul style="list-style-type: none"> <li>• Concentration: Signal Processing, Communications, and Controls</li> <li>• Thesis: <i>Synergistic Human-Machine Prediction: Active Error Analysis and Mitigation with Gaussian Process Regression</i> (advisor: Austin J. Brockmeier)</li> <li>• GPA: 3.93/4.00</li> </ul>                                                                                       | Newark, DE, USA<br>Aug 2018 – Aug 2020 |
| <b>Certificate Centro Psicopedagógico y de Investigación en Educación Superior (CEPIES)</b> , University Teaching                                                                                                                                                                                                                                                                                                                                                                     | La Paz, Bolivia<br>Aug 2015 – Nov 2015 |
| <b>B.S. Universidad Mayor de San Andrés (UMSA)</b> , Electronics Engineering<br><ul style="list-style-type: none"> <li>• Major: Control Systems</li> <li>• Graduation project: <i>Prototipo para generación de consignas de control a partir de los ritmos cerebrales alpha y mu. Caso de estudio: Factibilidad de uso de la plataforma Emotiv EPOC</i> (advisor: Javier Sanabria García)</li> <li>• Class valedictorian</li> </ul>                                                   | La Paz, Bolivia<br>Feb 2007 – Dec 2014 |

## Experience

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Computational Neural and Information Engineering Lab, University of Delaware</b> , Graduate Research Assistant<br><ul style="list-style-type: none"> <li>• Benchmarked out-of-distribution detection methods across CNN and ViT representations and training paradigms, and derived worst-case calibration error bounds under distribution shift.</li> <li>• Developed 3D convolutional networks and interpretation methods for brain age prediction from magnetic resonance elastography, in collaboration with Curtis L. Johnson's lab.</li> <li>• Built an interpretable post-concussion musculoskeletal injury risk model from longitudinal clinical data, in collaboration with Thomas A. Buckley's lab.</li> <li>• Ongoing: early prediction of cognitive decline on ADNI for the Delaware Longitudinal Study for Alzheimer's Prevention (Delaware Center for Cognitive Aging Research), and interpretable EEG biomarkers for epilepsy from routine scalp EEG with Nemours Children's Health.</li> </ul> | Newark, DE, USA<br>Jan 2022 – present<br>4 years 9 months  |
| <b>Computational Neural and Information Engineering Lab, University of Delaware</b> , Research Assistant<br><ul style="list-style-type: none"> <li>• Developed a synergistic system for human-machine prediction through active error analysis of a model's systematic errors using Gaussian process regression.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Newark, DE, USA<br>July 2019 – Aug 2020<br>1 year 2 months |

|                                                                                                                                                                                                                                                                                                                |                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Institute of Applied Electronics, UMSA</b> , Researcher <ul style="list-style-type: none"> <li>Designed and implemented embedded software for signal acquisition and control of a mobile robot prototype for multi-agent research.</li> </ul>                                                               | La Paz, Bolivia<br>June 2017 – Dec 2017<br>7 months |
| <b>Institute of Applied Electronics, UMSA</b> , Researcher <ul style="list-style-type: none"> <li>Developed a platform of mobile sensor nodes for controlled environments, covering the design of the robotic units and the data acquisition and signal processing tools.</li> </ul>                           | La Paz, Bolivia<br>Apr 2016 – Dec 2016<br>9 months  |
| <b>Agencia para el Desarrollo de la Sociedad de la Información en Bolivia (ADSIB)</b> , Research Intern <ul style="list-style-type: none"> <li>Designed and implemented an FPGA interface for Ethernet communication dedicated to packet acquisition and routing for the FPGA-based router project.</li> </ul> | La Paz, Bolivia<br>Sept 2015 – Dec 2015<br>4 months |
| <b>Digital Signal Processing Lab, Electronics Engineering Department, UMSA</b> , Laboratory Assistant <ul style="list-style-type: none"> <li>Responsible for the maintenance and operation of the digital signal processing laboratory.</li> </ul>                                                             | La Paz, Bolivia<br>July 2013 – Dec 2013<br>6 months |
| <b>Institute of Applied Electronics, UMSA</b> , Research Assistant <ul style="list-style-type: none"> <li>Developed a software application for EEG acquisition and spectral analysis via the fast Fourier transform.</li> </ul>                                                                                | La Paz, Bolivia<br>Mar 2012 – Aug 2012<br>6 months  |

## Publications

---

|                                                                                                                                                                                                                                                                                                                                                                                                  |          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Bounding Worst-Case Calibration Error in OOD Detection Under Distribution Shift</b><br>Claudio C. Claros-Olivares, Austin J. Brockmeier<br><a href="https://openreview.net/forum?id=aZnAoyzssl">openreview.net/forum?id=aZnAoyzssl</a> (AISTATS 2026 Workshop on Calibration for Modern AI)                                                                                                   | May 2026 |
| <b>A Systematic Analysis of Out-of-Distribution Detection Under Representation and Training Paradigm Shifts</b><br>Claudio César Claros Olivares, Austin J. Brockmeier<br><a href="https://arxiv.org/abs/2511.11934">arxiv.org/abs/2511.11934</a> (arXiv preprint arXiv:2511.11934 (under review))                                                                                               | Nov 2025 |
| <b>A Machine Learning Model for Post-Concussion Musculoskeletal Injury Risk in Collegiate Athletes</b><br>Claudio C. Claros, Melissa N. Anderson, Wei Qian, Austin J. Brockmeier, Thomas A. Buckley<br><a href="https://doi.org/10.1007/s40279-025-02196-4">10.1007/s40279-025-02196-4</a> (Sports Medicine, 55(8), 1971-1982)                                                                   | Jan 2025 |
| <b>MRI-based whole-brain elastography and volumetric measurements to predict brain age</b><br>Claudio C. Claros-Olivares, Rebecca G. Clements, Grace McIlvain, Curtis L. Johnson, Austin J. Brockmeier<br><a href="https://doi.org/10.1093/biomethods/bpae086">10.1093/biomethods/bpae086</a> (Biology Methods and Protocols, 10(1), bpae086)                                                    | Jan 2024 |
| <b>Integrative data analysis to identify persistent post-concussion deficits and subsequent musculoskeletal injury risk: project structure and methods</b><br>Melissa Anderson, Claudio Cesar Claros, Wei Qian, Austin Brockmeier, Thomas A. Buckley<br><a href="https://doi.org/10.1136/bmjsem-2023-001859">10.1136/bmjsem-2023-001859</a> (BMJ Open Sport & Exercise Medicine, 10(1), e001859) | Jan 2024 |
| <b>Mechanical Property Based Brain Age Prediction using Convolutional Neural Networks</b><br>Rebecca G. Clements, Claudio C. Claros-Olivares, Grace McIlvain, Austin J. Brockmeier, Curtis L. Johnson<br><a href="https://doi.org/10.1101/2023.02.12.528186">10.1101/2023.02.12.528186</a> (bioRxiv)                                                                                             | Jan 2023 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>A Machine Learning Approach for Modeling and Inverse Design of Polymer Blends</b><br>Soham Jariwala, Claudio César Claros Olivares, Zijie Wu<br><a href="http://bpb-us-w2.wpmucdn.com/sites.udel.edu/dist/d/12003/files/2022/05/Final_Dupont.postfeedback.pdf">bpb-us-w2.wpmucdn.com/sites.udel.edu/dist/d/12003/files/2022/05/Final_Dupont.postfeedback.pdf</a> (Project report, CHEG/CIS 867, University of Delaware (industry mentor: DuPont)) | May 2022 |
| <b>Identifying the Instances Associated with Distribution Shifts using the Max-Sliced Bures Divergence</b><br>Austin J. Brockmeier, Claudio César Claros Olivares, Matthew S. Emigh, Luis G. Sanchez Giraldo<br><a href="https://neurips.cc/virtual/2021/35481">neurips.cc/virtual/2021/35481</a> (NeurIPS 2021 Workshop on Distribution Shifts: Connecting Methods and Applications)                                                                | Dec 2021 |
| <b>Max-sliced Bures Distance for Interpreting Discrepancies</b><br>Austin J. Brockmeier, Claudio Cesar Claros, Carlos H. Mendoza-Cardenas, Yüksel Karahan, Matthew S. Emigh, Luis Gonzalo Sanchez Giraldo<br><a href="https://openreview.net/forum?id=D2Fp_qheYu">openreview.net/forum?id=D2Fp_qheYu</a> (Preprint, OpenReview)                                                                                                                      | Jan 2021 |
| <b>Synergistic human-machine prediction: Active error analysis and mitigation with Gaussian process regression</b><br>Claudio César Claros Olivares<br>(M.Sc. thesis, University of Delaware)                                                                                                                                                                                                                                                        | Jan 2020 |
| <b>Prototipo para generación de consignas de control a partir de los ritmos sensomotores. Caso de estudio: Factibilidad de empleo de la plataforma Emotiv Epoc</b><br>Claudio César Claros Olivares, Javier Sanabria García<br>(IEEE Bolivian Engineering and Technology Congress (BETCON), La Paz, Bolivia)                                                                                                                                         | Jan 2015 |

## Presentations

---

- **May 2026:** Poster, *Bounding Worst-Case Calibration Error in OOD Detection Under Distribution Shift*, AISTATS 2026 Workshop on Calibration for Modern AI, Tangier, Morocco.
- **May 2024:** Poster, ECE Research Day, University of Delaware. Best Poster Award, Signal Processing, Communication, and Control area.
- **Jul. 2024:** Co-author, *Predictors of Post-Concussion Musculoskeletal Injury Severity: An Integrated Data Analysis Approach*, UK Collaborating Centre on Injury and Illness Prevention in Sport (IOC Research Centre), Edinburgh, UK.
- **Jul. 2022:** Co-author, *Mechanical property-based brain age prediction using artificial neural networks*, World Congress of Biomechanics, Taipei, Taiwan.
- **Dec. 2021:** Poster, *Identifying the Instances Associated with Distribution Shifts using the Max-Sliced Bures Divergence*, NeurIPS 2021 Workshop on Distribution Shifts.
- **Feb. 2020:** Poster, *Active error analysis and mitigation for synergistic machine learning infrastructure*, DARWIN Computing Symposium, Data Science Institute, University of Delaware.
- **Nov. 2019:** Invited talk, *Learning an error-aware model for synergistic human-machine prediction*, SIG-AI Seminar, Department of Computer and Information Sciences, University of Delaware.
- **Apr. 2016:** Invited talk, *Digital signal processing algorithms and machine learning*, IEEE Seminar on Artificial Intelligence, IEEE Student Branch, Universidad Católica Boliviana, La Paz.
- **May 2015:** Invited talk on the *Sensor your Swarm* project, First International Scientific Congress on Engineering, La Paz.
- **Aug. 2014:** Invited talk, *Research on brain signal processing*, La Paz Ciudad Digital 2.0, representing the Institute of Applied Electronics, UMSA.

## Honors & Awards

---

- **2024:** Best Poster Award (Signal Processing, Communication, and Control area), ECE Research Day, University of Delaware.
- **2023:** Bendett Fellowship, Department of Electrical and Computer Engineering, University of Delaware.
- **2017:** Bolivian Government Scholarship for graduate studies abroad (30 of 100 scholarships awarded nationwide that year).
- **2015:** Honorable Mention, *Most Challenging Team*, Minesweepers: Towards a Landmine-Free World 2015, Antofagasta, Chile (Bolivian Team).
- **2015:** Senatorial Declaration, Chamber of Senators of the Plurinational Legislative Assembly of Bolivia, for outstanding participation in the NASA Space Apps Challenge 2015.
- **2015:** Resolution of the Honorable University Council, UMSA, for outstanding participation in the NASA Space Apps Challenge 2015.
- **2015:** Worldwide finalist, NASA Space Apps Challenge 2015: nationwide winner for *Best Robotics Theme Project* and worldwide finalist for *Best Mission Concept Project* with the project *Sensor your Swarm*.
- **2015:** Maximum score on the undergraduate graduation project, Electronics Engineering Department, UMSA.
- **2010:** Remarkable academic performance, UMSA. Congratulatory letter awarded to the best students of the year.

## Technical Skills

---

**Programming languages:** Python, C/C++, MATLAB, Octave, JavaScript, Julia

**Machine learning:** PyTorch, PyTorch Lightning, Hydra, TensorFlow, Keras, scikit-learn

**Scientific computing:** NumPy, SciPy, pandas, OpenCV, CUDA

## Languages

---

### Spanish

Native

### English

Fluent (TOEFL 104/120, ECCE)

### French

Intermediate (DELF B2)